

SVOBODNY, Russian Federation

WMO: 314450

Lat: 51.4555N

Lon: 128.1070E

Elev: 656

StdP: 14.35

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.5	-31.2	-40.3	0.6	-35.1	-36.2	0.7	-30.6	15.3	-0.6	14.1	-1.1	1.6	310	0.573

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	85.9	68.6	82.7	67.8	79.7	66.6	72.2	80.7	70.6	78.8	69.0	76.7	6.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.6	111.4	76.1	68.0	105.4	74.1	66.4	99.5	72.8	36.5	80.8	35.0	79.3	33.7	76.6	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
16.9	14.5	12.9	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-40.9	93.2	5.8	4.6	-45.0	96.5	-48.4	99.2	-51.6	101.8	-55.8	105.2
			-40.7	75.3	5.7	2.4	-44.8	77.0	-48.2	78.4	-51.4	79.7	-55.6	81.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	31.7	-10.5	-1.9	15.7	37.5	52.7	64.1	69.5	65.1	52.5	34.2	8.8	-9.7	(6)
(7)		DBStd	30.01	9.42	9.30	11.20	8.23	7.27	6.06	4.67	5.51	7.30	9.10	12.14	9.77	(7)
(8)		HDD50	8469	1875	1452	1064	385	54	1	0	1	58	493	1235	1851	(8)
(9)		HDD65	12450	2340	1872	1529	826	388	86	13	65	377	953	1685	2316	(9)
(10)		CDD50	1781	0	0	0	9	136	425	605	470	132	4	0	0	(10)
(11)		CDD65	289	0	0	0	0	5	60	153	69	2	0	0	0	(11)
(12)		CDH74	2527	0	0	0	14	131	665	1172	507	38	0	0	0	(12)
(13)		CDH80	643	0	0	0	4	27	205	311	94	2	0	0	0	(13)
(14)	Wind	WSAvg	5.2	4.4	4.9	6.0	6.9	6.3	4.4	4.0	4.2	5.0	5.9	5.2	4.6	(14)
(15)	Precipitation	PrecAvg	22.3	0.3	0.2	0.5	1.1	2.5	3.3	5.5	4.3	2.6	1.2	0.6	0.4	(15)
(16)		PrecMax	37.0	1.1	0.8	3.0	4.4	6.1	6.2	12.0	9.6	5.7	3.2	1.7	1.1	(16)
(17)		PrecMin	16.0	0.0	0.0	0.0	0.0	0.5	0.7	1.8	0.9	0.5	0.2	0.0	0.0	(17)
(18)		PrecStd	5.0	0.2	0.2	0.6	1.0	1.4	1.3	3.0	2.0	1.5	0.7	0.4	0.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.1	28.2	48.6	73.8	83.6	90.9	92.0	86.4	78.2	65.6	42.0	22.6	(19)
(20)		MCWB	13.1	23.6	36.8	53.5	59.5	68.0	71.4	69.8	63.8	51.1	35.5	19.8	(20)	
(21)		2%	DB	11.2	22.4	42.8	63.5	77.5	85.4	86.3	82.4	74.1	57.8	34.8	14.4	(21)
(22)		MCWB	9.0	18.8	33.6	46.6	56.7	66.2	70.5	69.0	61.6	46.1	30.0	12.3	(22)	
(23)		5%	DB	7.2	17.0	37.5	57.6	72.8	81.0	83.1	79.3	69.9	52.8	30.8	8.9	(23)
(24)		MCWB	5.4	14.2	30.3	43.7	54.4	65.1	69.6	67.6	58.4	43.3	27.7	7.2	(24)	
(25)		10%	DB	3.3	12.5	33.0	52.6	68.1	77.3	80.4	76.4	65.8	48.3	26.5	4.4	(25)
(26)		MCWB	1.9	10.1	27.9	40.4	52.4	63.5	68.4	66.3	56.5	40.5	23.4	3.0	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.6	23.9	37.9	54.1	61.8	71.9	74.8	73.6	66.6	53.5	35.8	19.8	(27)
(28)		MCDWB	15.8	27.5	47.4	72.4	77.9	86.4	83.0	80.7	74.2	63.4	41.7	22.4	(28)	
(29)		2%	WB	9.0	18.9	33.8	47.7	58.7	69.0	72.9	71.4	63.5	47.8	31.0	12.6	(29)
(30)		MCDWB	11.0	22.3	41.0	60.9	73.7	80.5	81.4	78.6	71.1	55.5	34.1	14.5	(30)	
(31)		5%	WB	5.4	14.2	31.0	44.4	56.3	66.8	71.4	69.5	60.8	44.3	27.7	7.3	(31)
(32)		MCDWB	7.0	17.0	37.2	56.1	68.8	77.4	79.9	76.4	67.5	51.7	30.8	8.9	(32)	
(33)		10%	WB	2.0	10.2	27.8	41.4	54.0	65.1	70.1	67.8	58.3	41.1	23.6	3.1	(33)
(34)		MCDWB	3.3	12.5	33.1	51.3	65.3	74.4	78.1	74.0	64.1	47.4	26.4	4.4	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	20.9	23.9	22.7	22.1	22.5	21.3	19.0	19.5	22.2	20.3	19.5	19.0	(35)
(36)		MCDBR	20.7	25.1	24.6	31.9	32.4	28.8	24.7	24.7	28.2	27.6	21.2	19.0	(36)	
(37)		MCWBR	18.8	21.9	16.9	18.6	14.9	12.4	10.9	12.0	15.4	17.1	16.7	17.3	(37)	
(38)		5% WB	MCDBR	20.7	25.1	23.4	29.3	26.9	24.0	20.2	19.1	24.0	24.5	21.0	18.7	(38)
(39)		MCWBR	18.7	21.9	16.2	17.6	13.8	11.7	10.3	10.3	14.0	16.0	16.8	17.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.264	0.298	0.376	0.553	0.617	0.499	0.480	0.423	0.392	0.388	0.314	0.268	(40)	
(41)		taud	2.150	2.075	1.911	1.669	1.620	2.000	2.109	2.259	2.260	2.094	2.120	2.099	(41)	
(42)		Ebn at Noon	234	256	251	217	211	243	246	255	249	216	206	207	(42)	
(43)		Edh at Noon	23	33	48	70	77	53	47	38	34	33	24	21	(43)	
(44)	All-Sky Solar Radiation	RadAvg	485	867	1389	1617	1719	1900	1736	1516	1187	782	494	375	(44)	
(45)		RadStd	17	24	70	167	134	215	139	100	97	43	26	20	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page